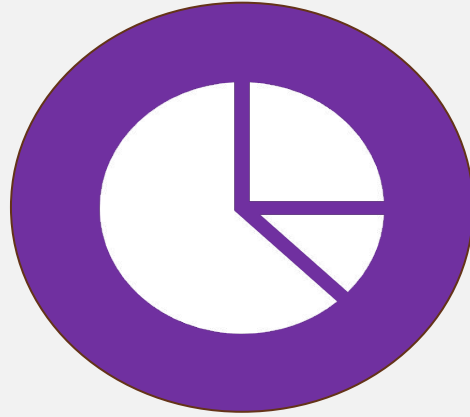




Initial Data Insights Presentation

Telecom User Data

Introduction

- ***Objective***

- ✓ To unlock and showcase key patterns and determine the value of further exploration in Telecom User Data

- ***Deliverables***

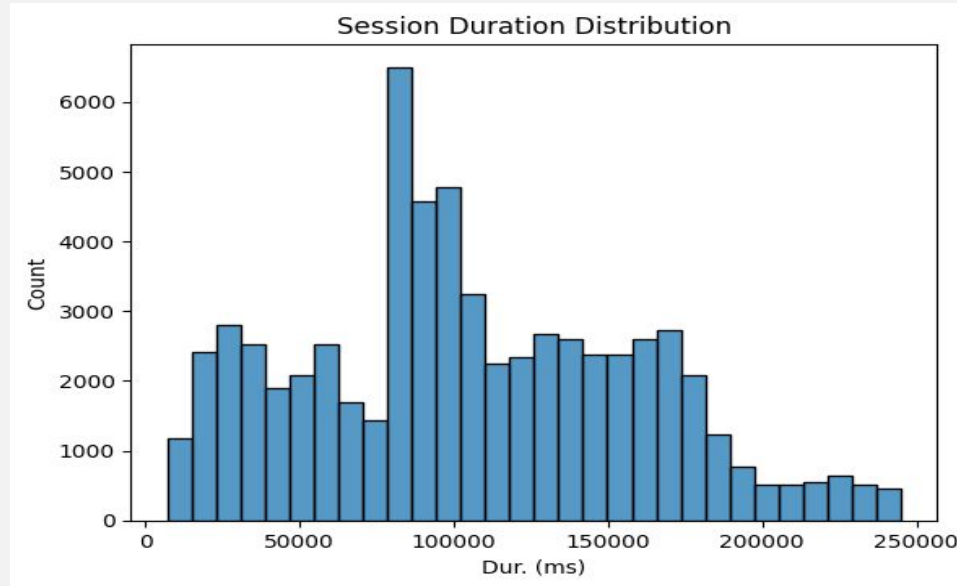
- ✓ Summary of insights, potential directions for investigation

Dataset Overview

- **55** columns and **150001** rows
- ***Key Columns:***
 - ✓ IMSI (user identifiers), MSISDN/Number, Session durations, Download/Upload volumes, Handset type and manufacturer
- ***High-Level Stats:***
 - ✓ Missing values in critical columns
 - IMSI (570), MSISDN(1066), Handset Type & Manufacturer (572)

Dataset Overview ...

- Distribution of numeric data (via a histogram)



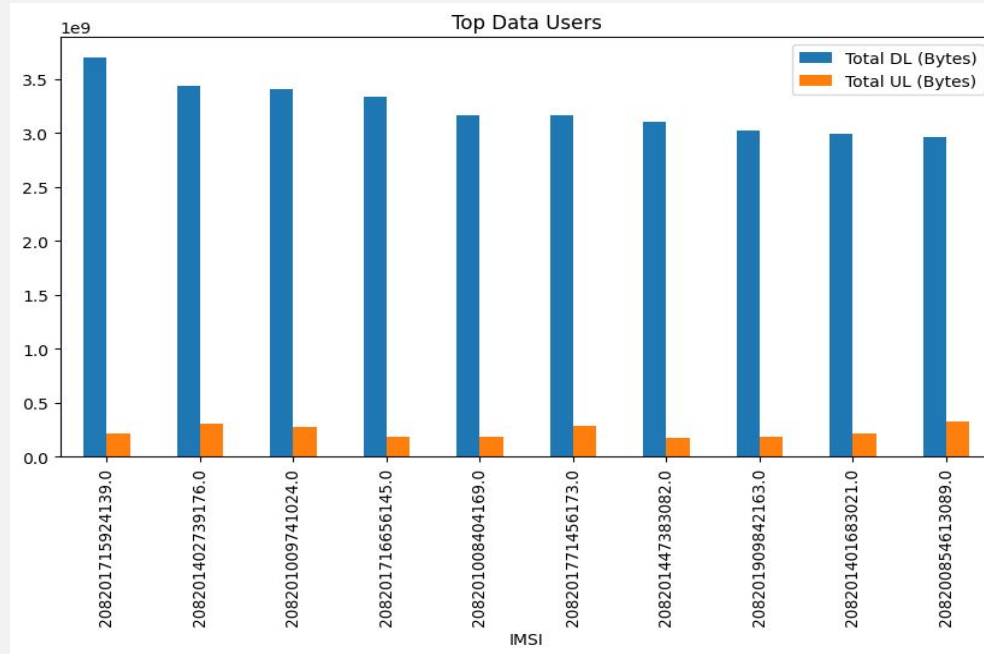
- Data appears rich with a mix of user behaviors, device data, and app usage insights, warranting deeper analysis

Unique User Insights

- Before cleaning:
 - Total Unique Users/IMSI before cleaning: 107265
 - Total Records: 150001
 - Average Records Per User: 1.40
- After cleaning:
 - Total Unique Users (IMSI) after cleaning: 54176
 - Total Records: 64848
 - Average Records Per User: 1.20

Top Data Users

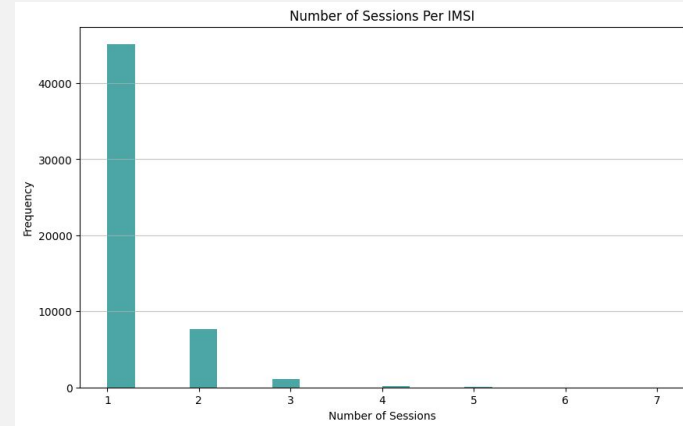
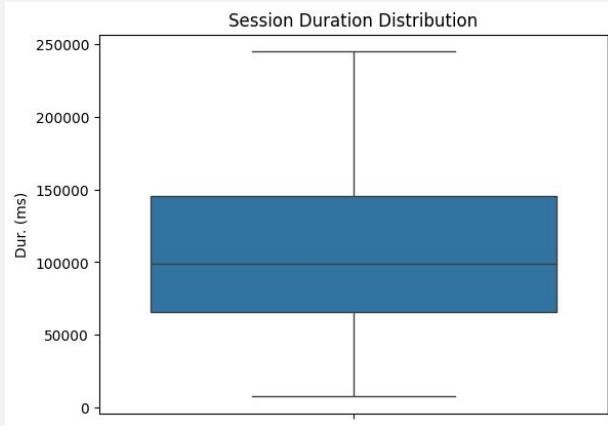
- Total UL/DL for top users:



- Concentrated behavior among high-use customers indicates potential segments for premium services or targeted ads

Average Session Duration

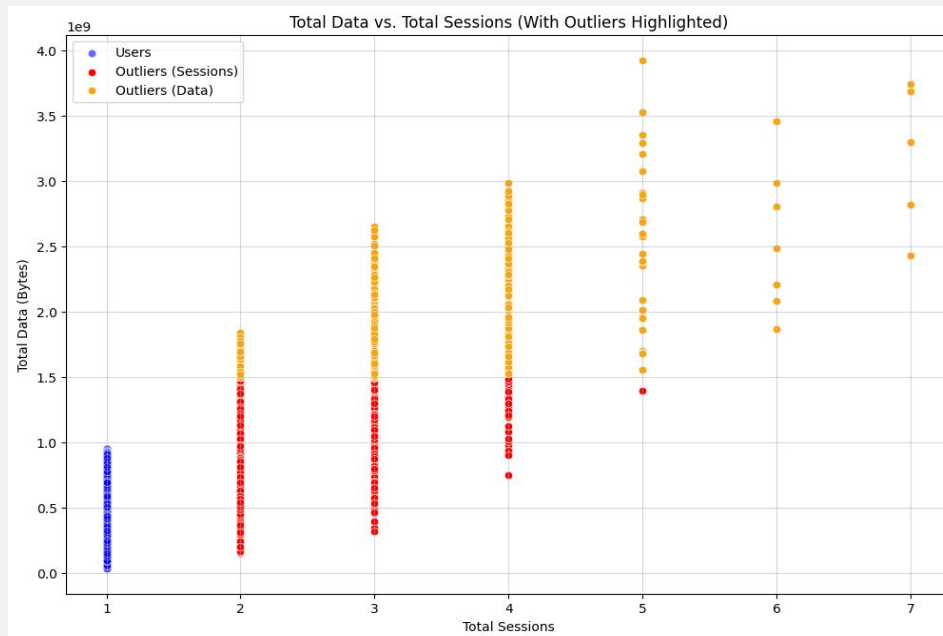
- Mean duration: **105645.63** milliseconds
- Distribution: Uneven with noticeable outliers



- A significant number of users have low activity, while a few users contribute disproportionately
- Users with long sessions may indicate potential retention strategies

Unusual Users Activities

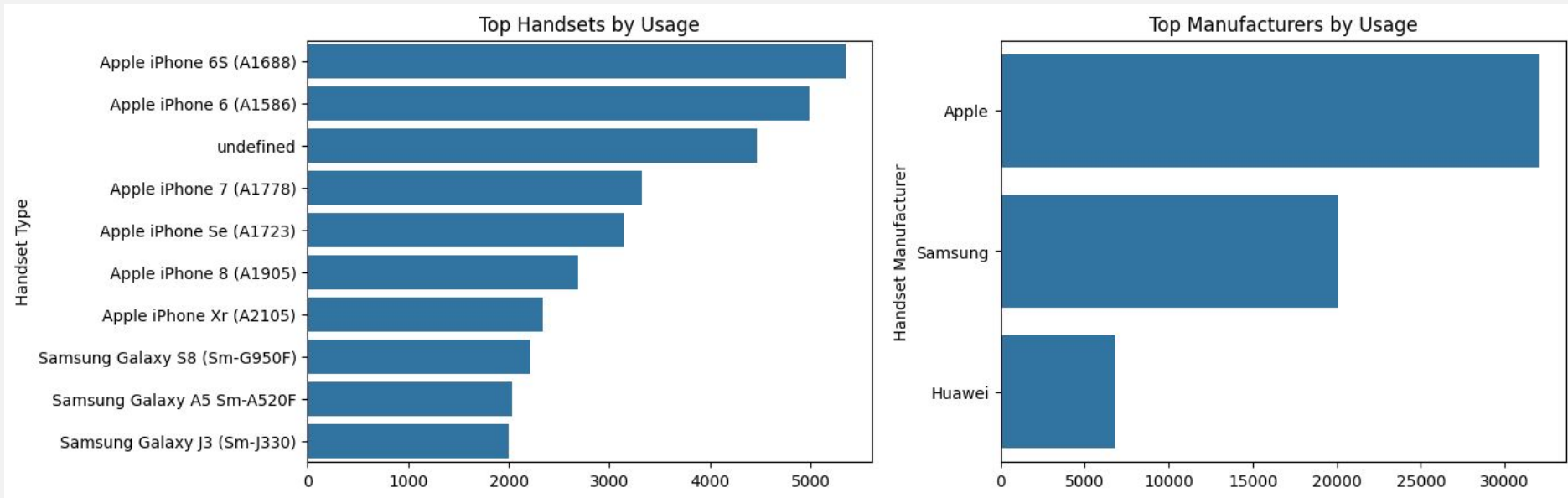
- Data & Session outlier users:



- Heavy users show a pattern of disproportionate data consumption—potential candidates for special customer retention programs

Handset Insights

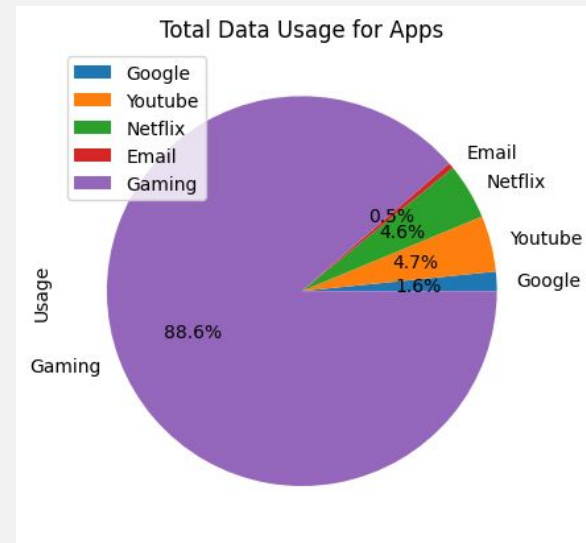
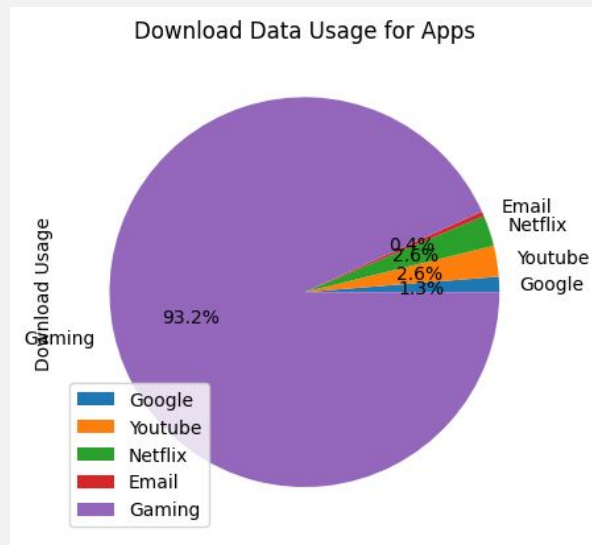
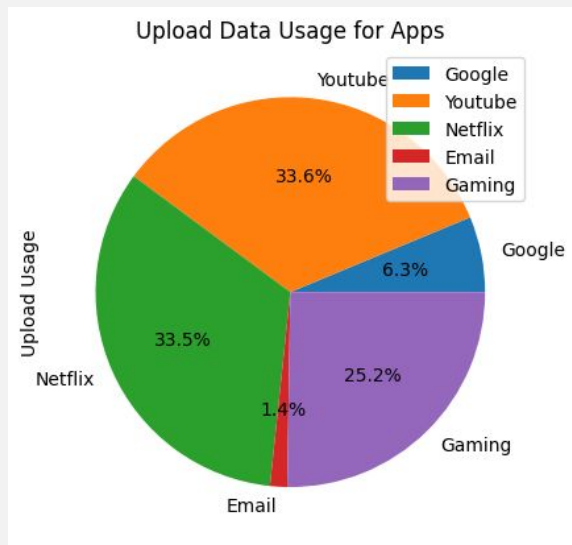
- Apple leads but secondary manufacturer, Samsung, has growing traction



- Device-specific behavior can guide marketing partnerships

Applications Data Consumption

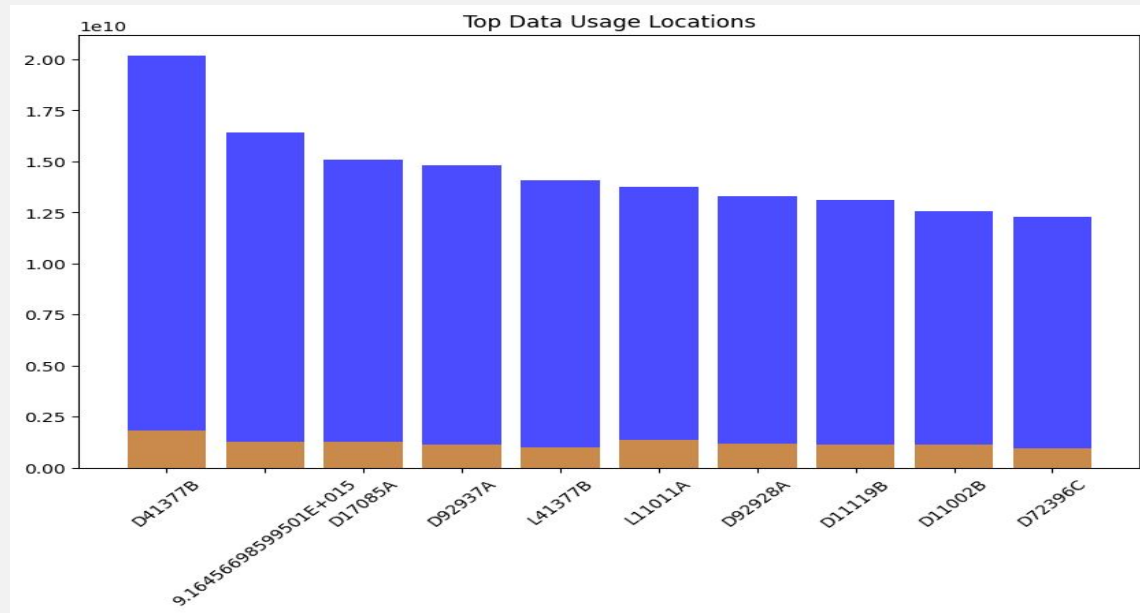
- **Gaming Apps:** dominate with 88.6% of total data consumption (93.2% downloads, 25.2% uploads)
- **Streaming Apps (YouTube, Netflix):** 5.2% of total data (66.7% of uploads)



- Top-used apps are clear targets for promotional tie-ins or network optimizations

Data Usage by Location

- Average duration comparison across geographies.



- Specific locations contribute significantly to session volumes—opportunities for infrastructure improvements

Recommendations for Further Investigation

- ✓ **Explore Device and App Relationships:** Does specific handset usage correlate with app preferences?
- ✓ **Dive Into Anomalous User Behavior:** Investigate heavy vs. dormant users for segmentation.
- ✓ **Geographic Optimization:** Target underperforming locations to enhance service quality.
- ✓ **Data Cleaning:** Address missing values for a more reliable baseline.

Final Takeaways

- **Key Findings:**

- Most activity is concentrated among a small subset of users, devices, and apps
- Clear trends for high-traffic locations and user behaviors

- **Actionable Next Steps:**

- Prioritize retention strategies for heavy users
- Optimize network services for top-used apps and geographies
- Consider partnerships with dominant handset manufacturers

➤ Initial analysis highlights valuable patterns; deep dives will likely yield strong ROI for further business actions